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Logic Dungeon II

THE SURVEYOR'S ROTUNDA

Purpose: Use the logic tools from Weeks 4–6 to decide which route is best supported as the safest path for twenty adventurers to the Upper Archive.

The Rotunda

Beyond the Hall of Inscriptions, the company of twenty stands in the **Surveyor's Rotunda**: a circular junction chamber with faded wall maps, a cracked compass rose on a central plinth, and water-stained note boards. Above them, in the cliffs of *Castrum Malae Ratiocinationis*, lies the Upper Archive and its original seals. Four routes leave the Rotunda. None looks obviously suicidal. All have surface appeal and surface red flags.

- **Raven Stair** — a carefully cut stone stair with carved ravens and recent shoring timbers; cold, dry air moves upward.
- **Well Passage** — a round opening with a steady draft of cold, damp air and a faint drip somewhere ahead.
- **Lantern Gallery** — a long passage with remnants of blue survey thread and old lantern brackets; the thread disappears at the first turn.
- **Rootbridge Path** — a lower, twisting route of stone arches over shallow depressions, patches of standing water, small floor pits, and clusters of cave insects. Scratched warnings urge travelers to turn back.

Earlier masons, imperial surveyors, maintenance crews, naturalists, and previous expeditions left notes of uneven quality. Some recorded careful observation; others recorded panic, haste, or unsupported inference. At each station the company must do two jobs: first, test the reasoning; second, decide what limited action the evidence justifies.

Two-test rule: *First apply the form test: if the premises were granted, would the conclusion follow? Then apply the fact test: are the premises established, doubtful, or false? The expedition needs the best-supported safe route, not a guarantee and not the route that merely looks finished.*

Tools: premises and conclusion; major, minor, and middle terms; practical syllogism map; A/E/I/O mood; figure; undistributed middle; two negative premises; validity; premise pressure point; soundness; form test vs. fact test.

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Part 1: The Raven Inscription

On a board near Raven Stair, an old scholarly hand has written:

All routes bearing an Archive seal are routes that reach the Upper Archive.

The Raven Stair is a route bearing an Archive seal.

Therefore, the Raven Stair is a route that reaches the Upper Archive.

A second scrap, in a different hand, notes recent tool marks and shoring on the lower landings but says nothing about the rock higher up.

Question 1.1

List the major premise, minor premise, and conclusion. Name the major, minor, and middle terms. Identify the mood and figure, then apply the form test.

Part 1 Q1

Question 1.2

Identify the pressure-point premise and one kind of evidence that would test it. What may the company record now, and should Raven be accepted for all twenty, rejected as impossible, or marked provisional?

Part 1 Q2

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Part 2: The Breath of the Well

A guide places a hand in the damp draft and argues:

All passages opening from the Upper Archive carry cold air.

The Well Passage carries cold air.

Therefore, the Well Passage opens from the Upper Archive.

Question 2.1

Let S = Well Passage, P = passages opening from the Upper Archive, and M = passages carrying cold air. Write the M/P/S pattern, identify mood and figure, and explain whether M connects S to P.

Part 2 Q1

Question 2.2

A scout can safely examine only the first turn. What observation may the company preserve, what has not been proved about destination or safety, and what limited investigation (if any) is justified?

Part 2 Q2

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Part 3: The Blue-Thread Rule

In the survey cabinet is a dated card still bearing the cabinet seal:

Record A — Chief Surveyor's Rule: *“Every route carrying an unbroken blue thread from this cabinet was inspected today and terminates in the Upper Archive.” The signature matches today's log.*

At Lantern Gallery the company can see blue thread entering the passage, but cannot yet tell which spool it came from or whether it remains unbroken beyond the first turn.

Question 3.1

Map the proposed Lantern argument as: All M are P; All S are M; therefore All S are P. Identify S, P, and M, the mood and figure, and the form-test result.

Part 3 Q1

Question 3.2

Apply the fact test. Which premise is established, which remains doubtful, and what is the present diagnosis? Choose and justify one action: reject Lantern, accept it for all twenty, or send two scouts only far enough to inspect the thread with a retreat line.

Part 3 Q2

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Part 4: Warnings at Rootbridge

Rootbridge looks the least inviting. Standing water lies in the low spots. Small pits mark the floors of the depressions. Insects cluster near the damp edges. Two older warnings are scratched into the arch stones.

Warning 1:

No passages with standing water are dry passages.

No Rootbridge low-spots are free of standing water.

Therefore, no Rootbridge low-spots are safe passages.

Question 4.1

State the mood and test the form. Explain what two negative premises fail to connect. What trap decision should the company make about assuming Rootbridge is unsafe from this warning alone?

Part 4 Q1

Warning 2:

All dangerous routes attract cave insects.

Rootbridge attracts cave insects.

Therefore, Rootbridge is a dangerous route.

Question 4.2

Identify the figure and form error. Explain why a true observation about insects would still not make this good proof, and state what limited action is justified at Rootbridge's mouth.

Part 4 Q2

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Part 5: Records That Change the Field

While the Lantern scouts are still out, the main company sorts three recovered documents from the Rotunda boards and a maintenance niche.

Record B — Mason's Ledger: *"Raven seals marked routes proposed for Archive access during the old construction. Some were completed; some were abandoned when the rock proved poor. The visible seals were never recut to show which. Weight on certain Raven landings may propagate old fracture lines."*

Record C — Cistern Report: *"The Well Passage crosses above an ice cistern. The cistern sends cold air and meltwater moisture into passages on both upper and lower levels. Recent tremors have increased saturation in the surrounding rock." The airflow reading beside Well is current.*

Record E — Gallery Amendment: *"Blue thread in Lantern Gallery was re-laid after a partial ceiling and floor collapse in the middle section. The unbroken thread reaches a landing. The inspection did not load-test the thinned floor where later drainage work was cut. Do not treat the thread as proof of full-company safety."*

Question 5.1

How does Record B change Raven's fact test without changing its form? Give Raven's present diagnosis and the company's route consequence.

Part 5 Q1

Question 5.2

What does Record C establish about Well, and why does it fail to repair the Well argument? Separately, how does Record E change Lantern's fact test even if the thread later proves unbroken to a landing? Give each route's present diagnosis and consequence.

Part 5 Q2

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Part 6: The Recovered Rootbridge Fragments

The Lantern scouts return after marking the middle gallery *structurally unproved*. Near Rootbridge, a retreat cord snags a loose panel. Behind it the company recovers three quieter records:

Record F — Mason’s Bridge Note: “Rootbridge arches were reinforced with iron pins. The pins remain sound. The visible floor pits are old shallow test pits or solution features, mostly 1–2 feet deep, bottoming on bedrock.”

Record G — Maintenance Comparison: “Standing water on Rootbridge is slow seepage from higher levels. Volume has remained roughly the same for decades. Old drains still function; the water does not accumulate or rise under ordinary conditions.”

Record H — Naturalist’s Field Card: “Vault striders (the small insects on Rootbridge) are harmless detritivores. They avoid movement, do not bite, and indicate stable humidity rather than active flooding.”

A practical syllogism is now proposed for Rootbridge:

All routes with sound iron-pin bridge reinforcement and long-stable low seepage, as confirmed by current mason and maintenance records, are routes safe for a full company of twenty.

Rootbridge is a route with sound iron-pin bridge reinforcement and long-stable low seepage, as confirmed by current mason and maintenance records.

Therefore, Rootbridge is a route safe for a full company of twenty.

Question 6.1

Rebuild the Rootbridge syllogism in standard order. Identify premises, conclusion, S/P/M, mood, figure, form judgment, and fact status under Records F and G. State what Record H contributes.

Part 6 Q1

Question 6.2

What new decision is justified now that was not justified in Part 4? Distinguish “safe under the supplied records” from “guaranteed free of every hazard,” and propose a limited-risk movement for the twenty adventurers.

Part 6 Q2

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Route	Form	Pressure point	Fact status	Diagnosis
Raven				
Well				
Lantern				
Rootbridge				

Complete the chart. Then identify the valid argument with a doubtful or undermined premise, the invalid argument containing a true observation, and the sound argument. Explain why these differ.

Which route is now best supported as the safest path for the twenty? Use at least three pieces of evidence from different parts and explain why each rejected route has weaker proof.

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Part 8: The Company's Decision

The stone guide declares: "Rootbridge looked ugly, so the early warnings must have been right to reject it. Lantern still has thread, so it must be safe. Validity settles everything." The company pauses before moving all twenty adventurers.

Question 8.1

Separate and repair the guide's claims. State what the sound Rootbridge argument establishes under the recovered records, what remains unguaranteed, and why appearance and unbroken thread are not by themselves enough.

Part 8 Q1

Question 8.2

Give the company's final movement plan for the safest best-supported path. Include one observation or safety check, one repair or retreat provision, and a 2–3 sentence explanation of why both the form test and the fact test were needed.

Part 8 Q2